



RAG and GraphRAG inside an agent loop

ADVANCED AGENTIC ARTIFICIAL INTELLIGENCE

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1. Why Retrieval-Augmented Generation Matters

LLMs face four key limitations:

- Hallucinations: Plausible but incorrect answers
- Outdated knowledge: Missing recent facts and policy changes
- Limited access: No visibility into private or domain-specific data
- Imprecise recall: Difficulty recalling details or citing sources

RAG retrieves relevant external evidence at inference time, grounding the model's response in up-to-date, specific information.

Key caveat: Retrieval can also hurt when sources are irrelevant, contradictory, or misleading. The real question is not whether to retrieve, but when retrieval improves accuracy-and when it does not.



2. The Core Question: When Does Retrieval Help?

Retrieval helps when:

- The answer requires recent, private, or domain-specific knowledge
- Retrieved passages are relevant, accurate, and sufficiently complete
- The task requires evidence or source grounding

Retrieval fails or hurts when:

- Evidence is incomplete, irrelevant, noisy, or contradictory
- The task requires synthesizing an entire corpus
- The model misinterprets evidence or treats weak sources as authoritative
- Irrelevant context distracts from a correct internal answer

Research question

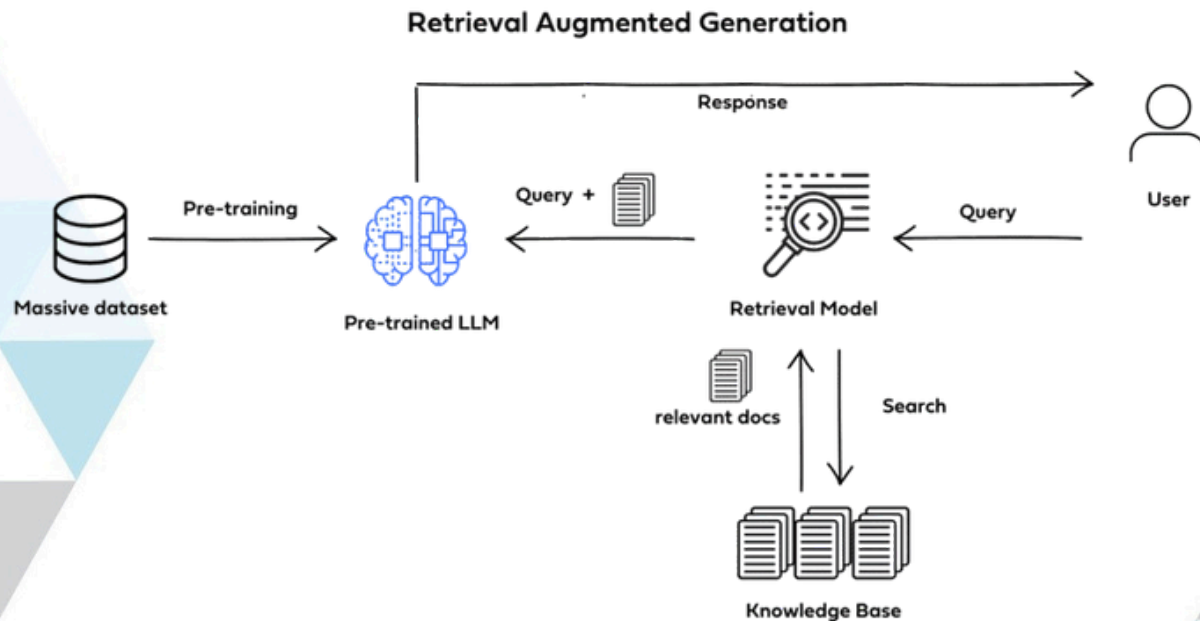
- How should an agent decide whether to retrieve, what to retrieve, and whether the evidence is useful enough to guide generation?

Papers

- Original RAG - Lewis et al.
- RAG survey - Gao et al.
- GraphRAG - Edge et al.
- Self-RAG - Asai et al.
- FLARE - Jiang et al.



3. Basic RAG Mechanism



Benefit: Relevant, trustworthy retrieval can improve accuracy, freshness, and traceability.

Risk: Irrelevant or misleading passages can make the answer worse.

2. From RAG Pipeline to Agent Decision Loop

Traditional RAG

- Query → Retrieve documents → Generate answer

Agent-based view

- Observe → Decide → Use a tool → Inspect results → Continue or answer

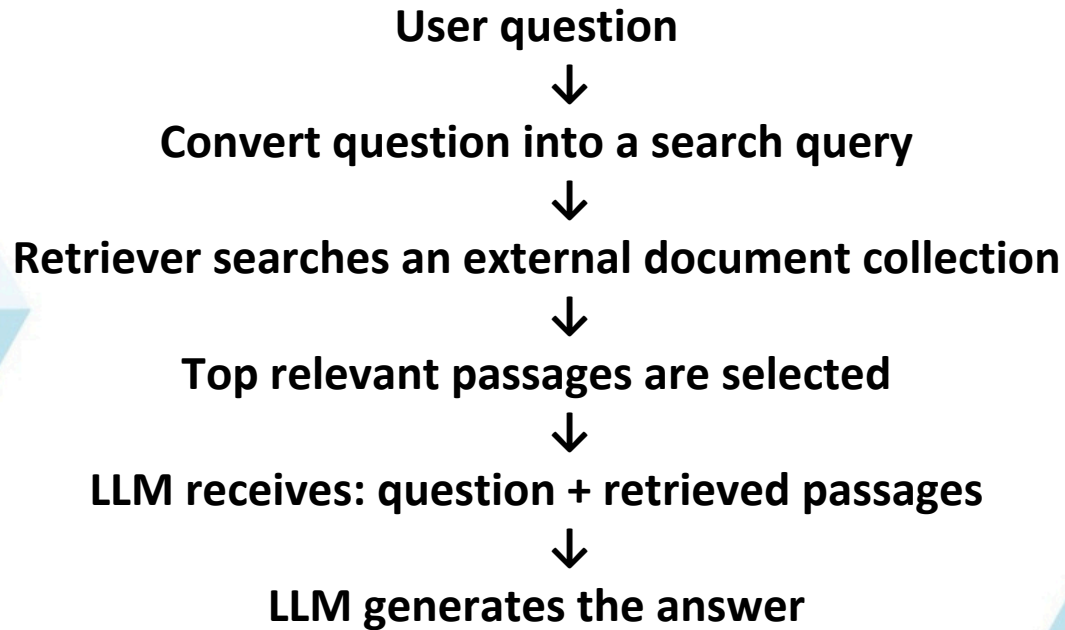
The agent may:

- Answer from existing knowledge
- Retrieve from documents, the web, or databases
- Refine the query and retrieve again
- Reject irrelevant or contradictory evidence
- Ask for clarification or stop when confident

“What passages should be retrieved?” → “Should the agent retrieve at all, which source should it use, and what should it do with the retrieved evidence?”

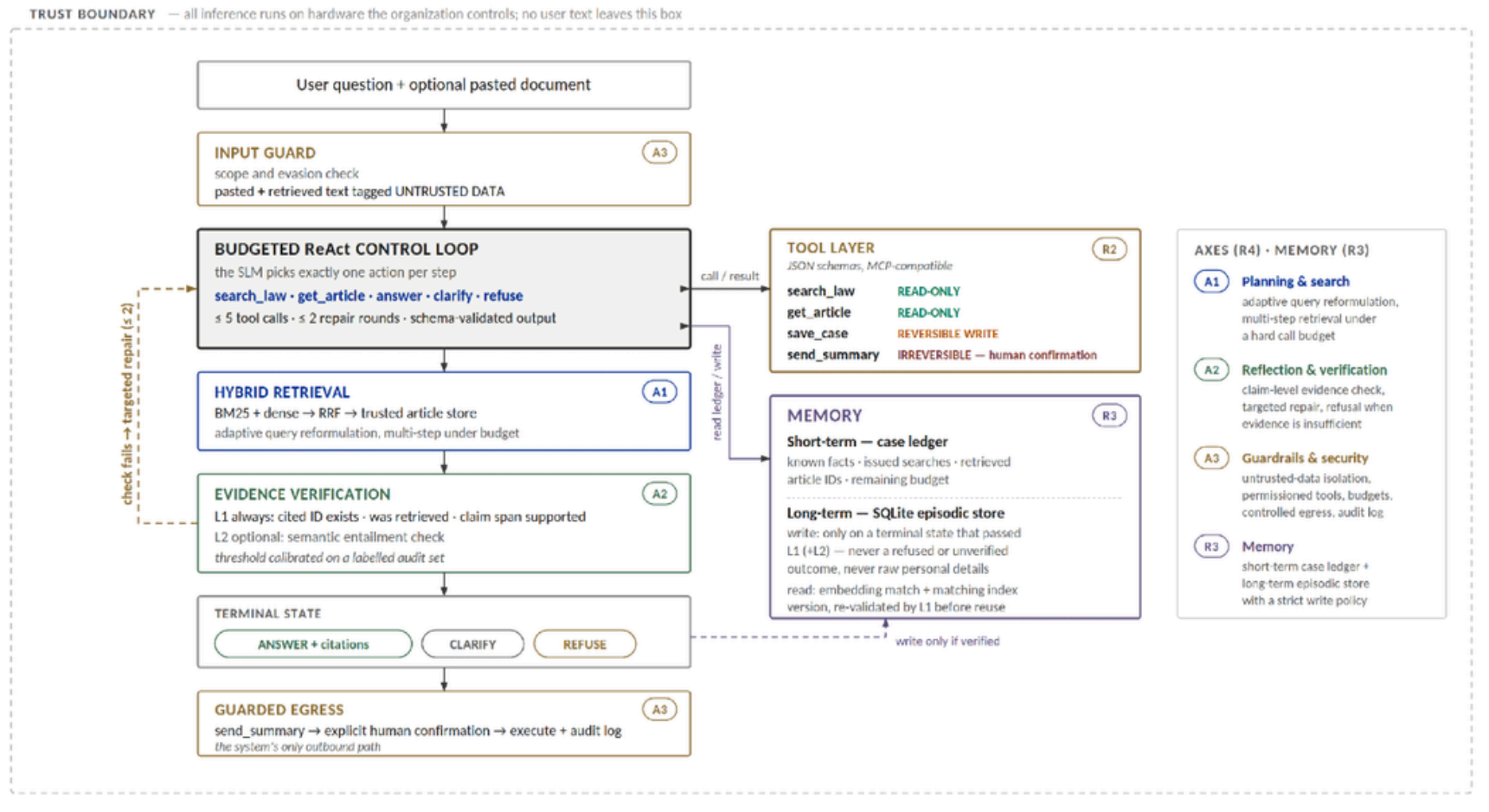
Retrieval can add missing evidence, but it also carries cost, noise, and risk of error.

4. Naive RAG

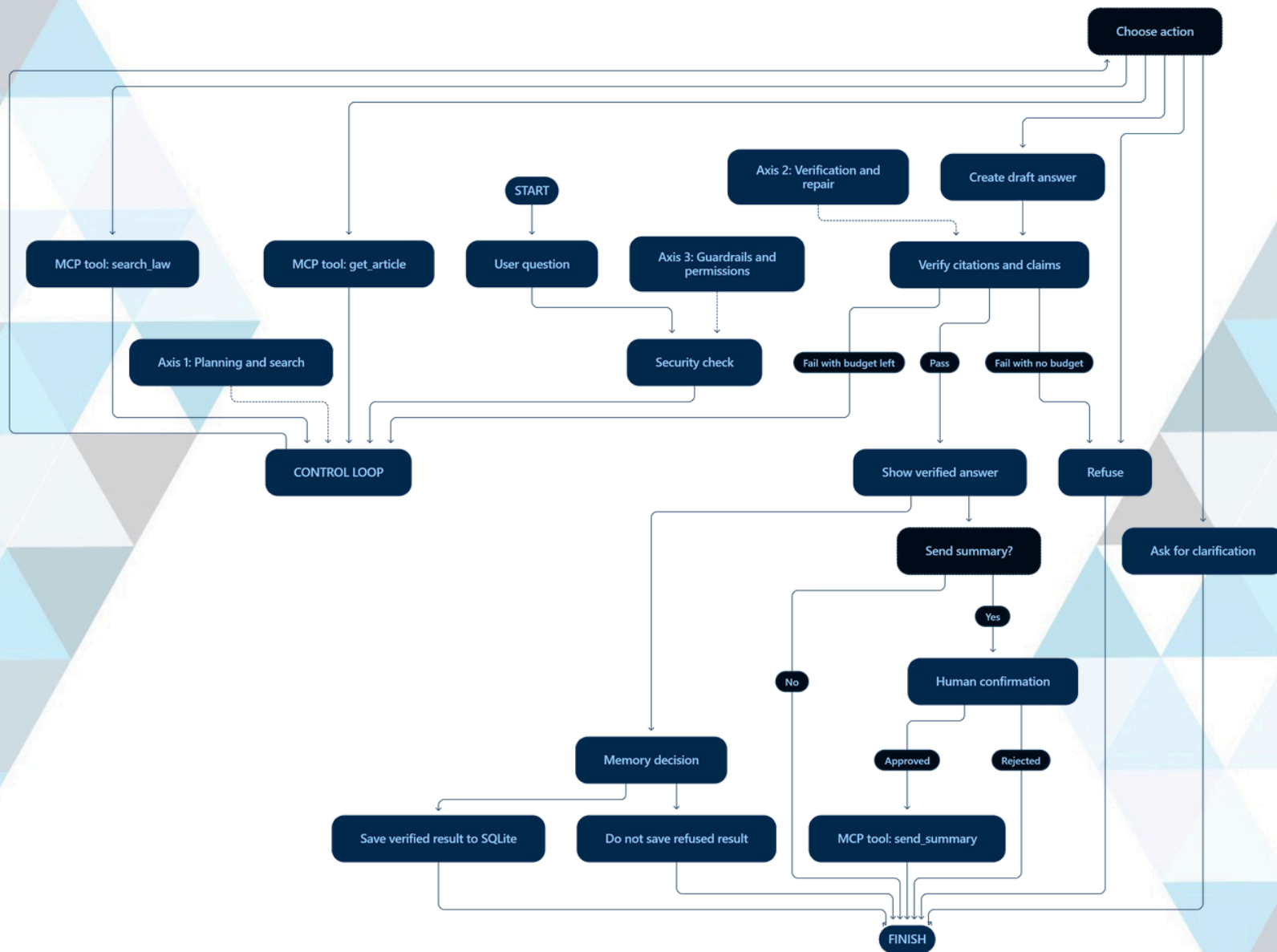


RAG retrieves evidence first, then conditions generation on that evidence.

2. Architecture



2. Architecture



3. Evaluation Plan

- Retrieval task set: 200 held-out VLQA query-to-article pairs
- End-to-end task set: 20 tasks
 - 10 single-provision tasks
 - 5 synthesis tasks
 - 3 unanswerable tasks
 - 2 adversarial tasks

Metrics:

- Retrieval: Recall@5, Recall@20, MRR
- Answer quality: unsupported-sentence rate, answer coverage
- Citation quality: citation precision and recall
- Safety: refusal correctness and attack-success rate
- Operations: p50/p95 latency and compute cost per task

Content

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- Active Retrieval and FLARE

When should the model retrieve?

FLARE direct predicts the next sentence before committing to it:

Draft next sentence → Check token probabilities →
Retrieve if any token is below threshold → Regenerate with evidence

- High confidence: keep the draft sentence.
- Low confidence: use the draft to search, then rewrite the sentence.

- ***Why Naive RAG Struggles with Global Questions***

- **Example question:** “What are the main themes in the dataset?”
- **Naive RAG:** Query → A few matching passages → Answer
- **The problem:** Matching passages may miss themes spread across the corpus.
- **Takeaway:** Global questions require corpus-wide synthesis.

- GraphRAG: From Documents to a Knowledge Graph

How GraphRAG works?

Documents → Graph → Communities → Summaries → Query-Focused Answer

- Extract entities & relationships
- Detect related communities
- Summarize each community
- Synthesize answers across communities

Key idea: Organize knowledge before answering.

- How GraphRAG Handles Global Questions

Before the question: Build hierarchical communities and summarize each one.

After the question:

- **MAP:** Community summaries → Partial answers
- **REDUCE:** Useful partial answers → Global answer

Example:

AI corpus → Generative AI | AI Chips | Regulation | Open Source



Major AI Trends

- RAG vs. FLARE vs. Self-RAG vs. GraphRAG

Method	Key idea	Main limitation	Best suited to
RAG	Adds knowledge from retrieved passages	Depends on retrieval quality	Knowledge-intensive tasks
FLARE direct	Retrieves when the next sentence contains a low-confidence token	May skip retrieval when wrong but confident	Long-form answers
Self-RAG	Decides when to retrieve and critiques evidence and output	Its decisions and critiques can be wrong	Adaptive retrieval
GraphRAG	Synthesizes community summaries across the corpus	Depends on graph and summary quality	Global questions

SUMMARY

- RAG: Retrieve, then generate
- FLARE direct: Retrieve when the next sentence is uncertain
- Self-RAG: Decide whether to retrieve, then critique
- GraphRAG: Summarize communities for global questions

- What Does Each Method Assume?

Each method depends on the quality of its decision signal or knowledge representation.

Method	Key assumption
RAG	The retriever finds useful passages
FLARE direct	Low token confidence signals a need for evidence
Self-RAG	The model can judge when to retrieve and whether evidence supports its answer
GraphRAG	The graph and community summaries represent the corpus well

- Evaluation Weaknesses

Good benchmark $\neq$ Good everywhere. High benchmark scores leave four questions open:

- Benchmark dependence: Does the result hold on other tasks?
- Limited domains: Does it generalize to other document collections?
- Faithfulness: Does the evidence support the answer?
- Retrieval decisions: Was retrieval needed at that point?

Example: Correct answer $\neq$ Retrieval was necessary

- **When More Context Makes Things Worse**
- More context does not guarantee better evidence.
- More retrieved text can introduce:
 - Irrelevant or distracting passages
 - Conflicting evidence
 - Important facts buried in excess context
 - Highly ranked but misleading documents
- **Result:** The answer may become less accurate.

- Concrete Case: Retrieval Can Turn a Correct Answer into a Wrong One

Question: Which university did George H. W. Bush attend? Correct answer: Yale University.

- GPT-3.5 without retrieval: “Yale University” → Correct
- GPT-3.5 with a BM25-retrieved passage: The passage says that “the George H. W. Bush Presidential Library is located on the Texas A&M University campus”. The model answers “Texas A&M University” → Incorrect

- What the Papers Do NOT Show

Avoid conclusions beyond the evidence

- Retrieval $\neq$ Faithful grounding
- High confidence $\neq$ Factual correctness
- Knowledge graph $\neq$ Verified facts
- Benchmark gains $\neq$ Effective agent decisions

Content

1. Demo

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